Cardinality of Infinite Sets

- **There are many infinities!**
- **Definition:** Two infinite sets A and B have the same cardinality if and only if there is a one-to-one correspondence from A to B.
- **Definition:**

  $$|\mathbf{N}| = \aleph_0$$

  where $\mathbf{N}$ is the set of natural numbers {0, 1, 2, …}
- An infinite set S with $|S| = \aleph_0$ is called is **countable.**
  - Set S is countable if and only if we can list its elements in sequence indexed by the natural numbers.
  - It is possible that $|S| = \aleph_0$ and $S \subset \mathbf{N}$ (e.g., set of positive even numbers $\subset \mathbf{N}$), or $\mathbf{N} \subset S$ (e.g., $\mathbf{N} \subset \mathbf{Z}$)
- An infinite set that is not countable is **uncountable.**
  - Which set has higher cardinality: $\mathbf{N}$ or the set of real numbers between 0 and 1?

Hilbert's Grand Hotel

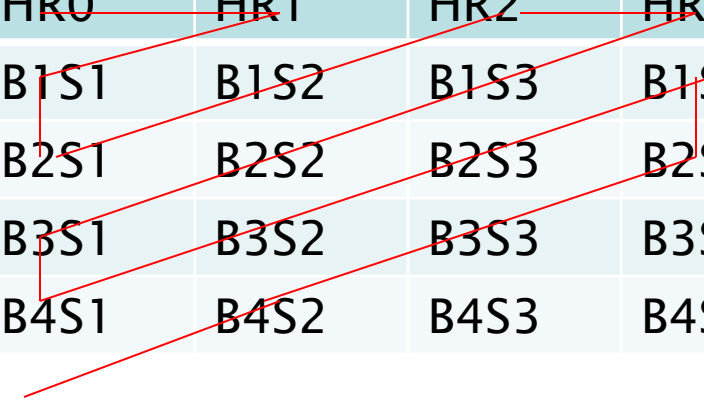
- Assume a hotel with infinite number of rooms, 0, 1,2,…, all full. Guest x is initially in room $r(x)=x$.
- A new guest arrives. How to accommodate her?
  - Move each existing guest to the next room $r(x) = x + 1$, so that the first room becomes available. Put new guest in Room 0.
- A bus with a finite number (e.g., 100) of new guests arrives. How to accommodate them?
  - Move each existing guest to room $r(x) = x + 100$, so that the first 100 rooms become available.
- A bus with an infinite but countable number of new guests arrives. How to accommodate them?
  - Move each existing guest to room $r(x) = 2x$, so that all odd numbered rooms become available.

https://www.youtube.com/watch?v=OxGsU8oIWjY&ab_channel=Veritasium

Hilbert's Grand Hotel (cont)

- Assume a hotel with infinite number of rooms, 0, 1,2,… all full. Guest x is initially in room $r(x)=x$.
- An infinite countable number of buses, each with an infinite countable number of new guests arrives. How to accommodate them?

Hotel	HR0	HR1	HR2	HR3
Bus 1	B1S1	B1S2	B1S3	B1S4
Bus 2	B2S1	B2S2	B2S3	B2S4
Bus 3	B3S1	B3S2	B3S3	B3S4
Bus 4	B4S1	B4S2	B4S3	B4S4



- Create a room sequence by following the red line: HR0, HR1, B1S1, B2S1, B1S2, HR2, HR3, …

Hilbert's Grand Hotel (cont)

- Assume a hotel with infinite number of rooms, 0,1,2,… all full. Guest x is initially in room $r(x)=x$.
- An (uncountable) number of guests arrives. Each guest has a unique ID which is a string of 0,1. How to accommodate them?
- Assume that you accommodate guests in rooms in increasing order of ID.

Room	Guest
Room 1	000011111..
Room 2	010001111..
Room 3	011001010
…	…

- I generate a guest ID 100…, by flipping the first bit of the first guest, the second bit of the second guest etc.
  - This new ID 100… is different from every other on (at least) the diagonal, i.e., different from ID1 on first bit, from ID2 on second bit etc.
  - No matter what I do, I can always find a guest that cannot be accommodated.

Example of Countable Sets

- To show that an infinite set S is countable, we must find a one-to-one correspondence between $\mathbf{N}$ and S.
- $\mathbf{Z}^+ = \{1, 2, 3, \ldots\}$.
  Let $f: \mathbf{N} \to \mathbf{Z}^+$ be $f(x) = x + 1$

N	0	1	2	3	..
	↓	↓	↓	↓	
Z^+	1	2	3	4	..

- The set of nonnegative even numbers.

$$f(x) = 2x$$

N	0	1	2	3	..
	↓	↓	↓	↓	
even Z^+	0	2	4	6	..

- The set of all integers $\mathbf{Z}$.
- List all integers as: $0, 1, -1, 2, -2, 3, -3,$ ...

Define $f(x) = \begin{cases} -\frac{x}{2}, if\ x\ is\ even \\ \frac{x+1}{2}, if\ x\ is\ odd \end{cases}$

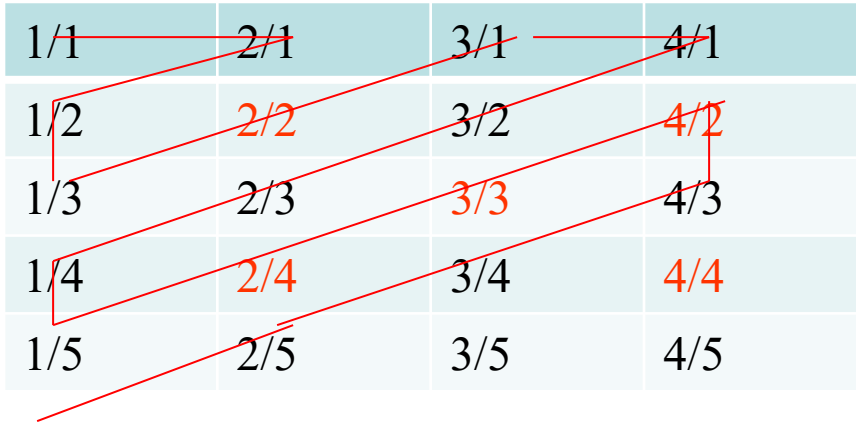
N	0	1	2	3	4	..
	↓	↓	↓	↓		
Z	0	1	-1	2	-2	..

Countable and Uncountable Sets

- Countable:
  - The set of odd numbers
  - The set of even numbers
  - The set of integers
- Uncountable:
  - The set of real numbers
- The continuum hypothesis: There is no set whose cardinality is strictly between that of natural numbers and that of real numbers (Hilbert's first problem).

Set of Positive Rational Numbers

- Every positive rational number can be written as the quotient p/q of two positive integers
- Arrange rational numbers in 2D array, where the first row contains numbers with q=1, the second row q=2, etc. Within the same row, arrange in increasing order of p.



1/1	2/1	3/1	4/1
1/2	2/2	3/2	4/2
1/3	2/3	3/3	4/3
1/4	2/4	3/4	4/4
1/5	2/5	3/5	4/5

- Red line lists rational numbers in increasing order of $p+q$: 1/1, 2/1, 1/2, 1/3, 2/2, 3/1 etc. Do not list duplicates.
- Is the set of *all* rational numbers countable?
  - Yes: If A and B are countable, then $A \cup B$ is countable

Set of Real Numbers Between 0 and 1

- Assume that this set is countable. Then, the set of all real numbers between 0 and 1 can be listed in some order

r_1=0.$d_{11}d_{12}d_{12}$ ..

r_2=0.$d_{21}d_{22}d_{23}$ ..

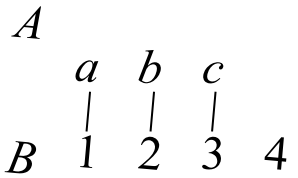
r_3=0.$d_{31}d_{32}d_{33}$ ..

…, where d_{ij} is a digit between 0 and 9.

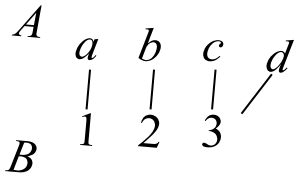
- I generate a real number r=0.$d_1d_2d_3$.. such that $d_1 \neq d_{11}, d_2 \neq d_{22}, d_3 \neq d_{33}$..
- r is different from each r_i on digit d_i, contradicting the assumption that the set of all real numbers between 0 and 1 can be listed in some order.
- Thus, this set is uncountable.
- This type of proof is called Cantor diagonalization argument

Recap on Functions

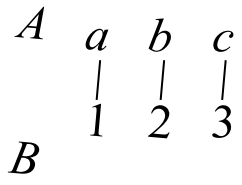
A function $f:A \to B$ is **injection** (or **one-to-one**) if and only if $f(a) = f(b)$ implies that $a = b$ for all a and b in the domain of f.



A function $f:A \to B$ is **surjection** (or **onto**) if and only if for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$.



A function $f:A \to B$ is a **bijection** (or *one-to-one correspondence*) iff it is **both one-to-one and onto**.



Recap on Countable Sets

- A infinite set S with cardinality $|S| = |\mathbf{N}| = \aleph_0$ is called is **countable**, where $\mathbf{N}$ is the set of natural numbers {0, 1, 2, …}
- S is countable, if there is bijection $f: \mathbf{N} \to S$
- Prove that the set E of nonnegative even numbers is countable.
- Define function $f(a) = 2a$, $\mathbf{N} \to E$ and show that it is bijection.
- Assume $f(a) = f(b)$. Then, $2a = 2b \Rightarrow a = b$. Thus, f injection.
- Assume any number $a' \in E$. Since a' is even: $\exists k \in N\ a' = 2k = f(k)$. Thus, f surjection.
- Thus, f bijection.
- Countable Sets: Integers, Rational Numbers
- Uncountable Sets: Real Numbers

Properties of Functions and Cardinality of Infinite Sets

For two sets A and B

- If there is bijection $f: A \to B$, then $|A| = |B|$ (**Bijection principle**)
- If there is injection $f: A \to B$, then $|A| \leq |B|$
- If there is injection $f: A \to B$, but not bijection, then $|A| < |B|$

Schröder-Bernstein Theorem: if $|A| \leq |B|$ and $|B| \leq |A|$, then $|A| = |B|$.

- Example: Show that $|(0,1)| = |(0,1]|$
  - First interval contains all real numbers x: $0<x<1$
  - Second interval contains all real numbers x: $0<x\leq 1$
  - Not easy to find a bijection between the two sets. Easier to find two injections and apply the theorem.
- An injection $(0,1) \to (0,1]$: $f(x) = x$. Thus, $|(0,1)| \leq |(0,1]|$
- An injection $(0,1] \to (0,1)$: $f(x) = x/2$. Thus, $|(0,1]| \leq |(0,1)|$
- Thus, according to theorem: $|(0,1)| = |(0,1]|$

Additional Theorems

- Theorem. The cardinality of any set S is lower than the cardinality of its powerset $P(S)$: $|S| < |P(S)|$
  - It holds for both finite and infinite sets
  - $P(\mathbf{R})$ has higher cardinality than the set of real numbers $\mathbf{R}$

- Definition: A function is computable if there is a computer program that finds the value of the function on any input. Otherwise, it is uncomputable.
- Theorem: There exist functions from $\mathbf{N}$ to {0, 1} that are uncomputable.
  - Intuition: The set of computer programs is countable, while the set of functions is uncountable.

- Halting Problem (Alan Turing): Is there a program that determines whether another program terminates for some input?
  - There is no general algorithm that solves the halting problem.